INDEPENDENT UNIVERSITY, BANGLADESH

SCHOOL OF ENGINEERING, TECHNOLOGY AND SCIENCES

Department of Computer Science and Engineering

A Project Report On

Student on Duty (SoD) Management System

An Automated Enterprise Platform for Timetable Parsing, Wireless RFID Attendance Tracking, and Streamed Payroll Claims

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Abstract – The operational management of Student Assistant (SA) duty assignments, academic course timetable conflicts, shift attendance tracking, and monthly payroll claims within university academic departments has traditionally suffered from high administrative friction, manual cross-referencing errors, unverified duty presence, and billing reconciliation disputes. This report presents the **Departmental Student on Duty (SoD) Management System**, a production-ready enterprise platform built with an asynchronous FastAPI Python backend, a React 18 TypeScript Single Page Application (SPA), and PostgreSQL relational persistence. Executed across four structured Agile development sprints (v1.0.0 through v4.0.0), the system integrates automated regular expression (Regex) IRAS academic schedule parsing, wireless 13.56 MHz RFID hardware card attendance tracking, a standalone Light-Mode RFID Kiosk terminal (/manager/attendance/duty/1/kiosk), an automated broadcast proxy shift-swap matching engine, and a streamed CSV payroll processing pipeline (app/services/payroll.py). The platform strictly complies with Model-View-Controller (MVC) decoupling standards, shielding presentation pages through custom React hooks (useSchedule.ts, useSwaps.ts, useAttendance.ts). Comprehensive SDLC verification links user stories (US-001 to US-013) to automated pytest test suites under backend/tests/, achieving a 100% pass rate across all 16 execution cases.

1 Introduction

University academic departments rely heavily on Student Assistants (SAs) to supervise computer laboratories, assist faculty members during lecture sessions, maintain hardware equipment, and invigilate midterm and final examinations. Managing student duty operations requires coordinating academic course timetables, verifying physical presence during assigned duty hours, managing sudden shift-swap requests, and processing monthly billing claims.

Historically, the Department of Computer Science and Engineering at Independent University, Bangladesh (IUB) managed these workflows through fragmented Google Sheets, manual messaging threads, and paper timesheets. This led to frequent schedule overlaps with student class hours, missed duty coverage, unresolved shift swaps, and billing disputes during monthly payroll processing.

1.1 Current vs. Proposed Process

Figure 1: Current Manual SA Duty Management Process

Figure 2: Proposed Automated SoD Management Process

1.2 Agile Release Roadmap

To systematically address these operational bottlenecks, the software engineering project was executed across 4 structured Agile development sprints, culminating in the production-ready v4.0.0 final release.

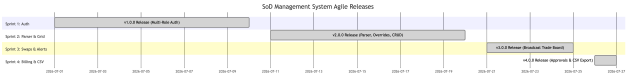

Figure 3: Agile Development Release Roadmap (Sprints 1 through 4)

1.3 Project Objectives

The primary objective of the SoD Management System is to deliver a centralized, secure, and scalable web application that automates duty scheduling, shift verification, and financial claims. Specific technical goals include:

1. **Automated Timetable Parsing:** Parse raw IRAS academic course schedules into conflict-free student availability matrices.
2. **Wireless RFID Attendance & Kiosk:** Implement real-time check-in and check-out tracking using physical 13.56 MHz RFID cards and a standalone Light-Mode RFID Kiosk subpage.
3. **Automated Shift Swaps:** Provide an intelligent broadcast proxy engine that matches swap requests against real-time student availability.
4. **Payroll Stream & Approval Pipeline:** Streamline monthly bill claim submissions, multi-tier approvals, and dynamic CSV payroll streaming.

2 Problem Definition & Stakeholder Analysis

The operational challenges in managing Student Assistants stem from information silos and manual record-keeping.

2.1 Operational Bottlenecks

- **Manual Availability Verification:** Comparing student class schedules against lab duty slots required hours of manual cross-referencing.
- **Unverified Duty Attendance:** Traditional sign-in sheets were susceptible to proxy attendance and unverified timestamps.
- **Payroll Reconciliation Errors:** Paper billing claims often contained arithmetic miscalculations or unverified duty hours.

2.2 Stakeholder Analysis

Table 1: Stakeholder Matrix and Core Requirements

Stakeholder	Core Requirements & Responsibilities
Student	Timetable parsing, shift swap requests, check-in/out, bill claim submission.
Faculty	Task creation, assistant oversight, duty verification.
Lab Manager	Shift slot creation, RFID kiosk operation, attendance roster audits.
Dept Manager	Departmental duty allocation, budget oversight, CSV payroll stream export.

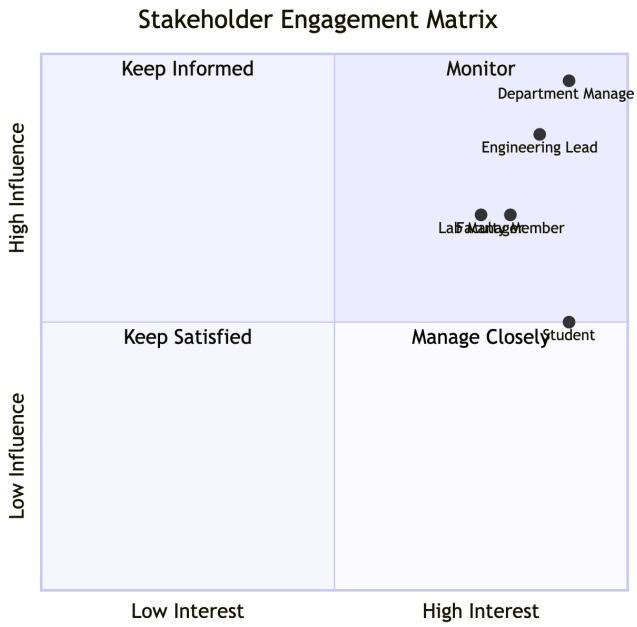

Figure 4: Stakeholder Analysis & Influence Matrix

2.3 Information Gathering Workflow

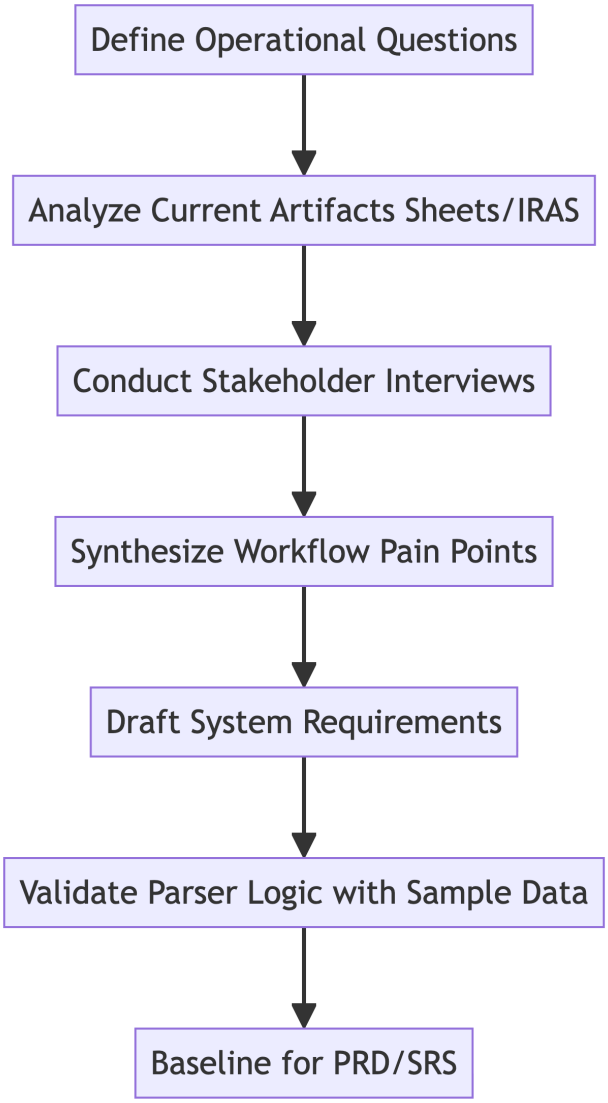

Figure 5: Information Gathering & Elicitation Workflow

2.4 Feasibility Analysis

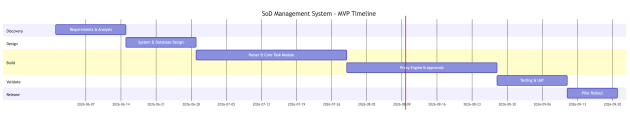

Figure 6: Technical & Operational Feasibility Decision Tree

2.5 User Journey Mapping

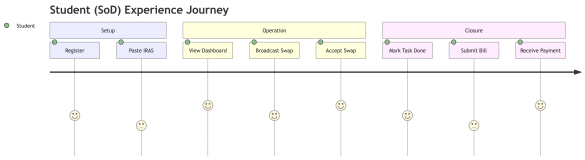

Figure 7: User Journey Map: Student & Manager Dual Perspective

3 Literature Review & Comparative Analysis

Existing commercial workforce management tools (e.g., Handshake, Shiftboard) target general corporate environments and lack integration with university academic course registration systems such as IRAS.

Table 2: Comparative Matrix: Manual vs. Generic vs. SoD System

Feature	Manual	Generic Tools	SoD System
IRAS Schedule OCR/Parser	No	No	**Yes**
RFID Hardware Kiosk	No	Third-party	**Integrated**
Broadcast Proxy Swaps	No	No	**Automated**
Role Payroll CSV Stream	No	Generic	**Automated**
MVC Layer Decoupling	N/A	Proprietary	**Compliant**

4 System Architecture & Design Models

The system adopts a 3-tier architecture adhering strictly to Model-View-Controller (MVC) standards.

4.1 Data Flow Diagrams (DFD)

The system's data flows are modeled across three decomposition levels, progressing from context-level interactions to subsystem-specific processing.

Figure 8: DFD Context Diagram (Level 0)

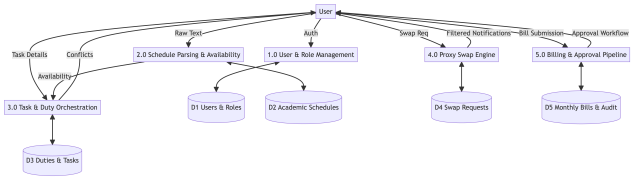

Figure 9: DFD Level 1: System Decomposition

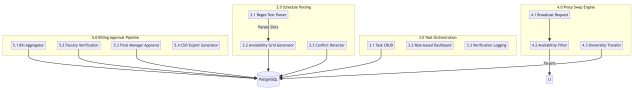

Figure 10: DFD Level 2: Shift Swap Subsystem

4.2 MVC System Conceptual Architecture

To eliminate monolith clutter, business logic is isolated into specialized backend service modules and custom React hooks.

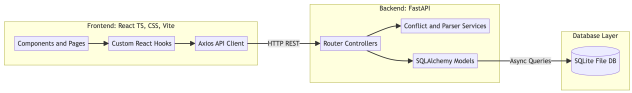

Figure 11: Decoupled 3-Tier MVC System Architecture

4.3 Database Entity Relationship (ER) Model

The database schemas are built using SQLAlchemy ORM entities, maintaining relational integrity across student duty cycles.

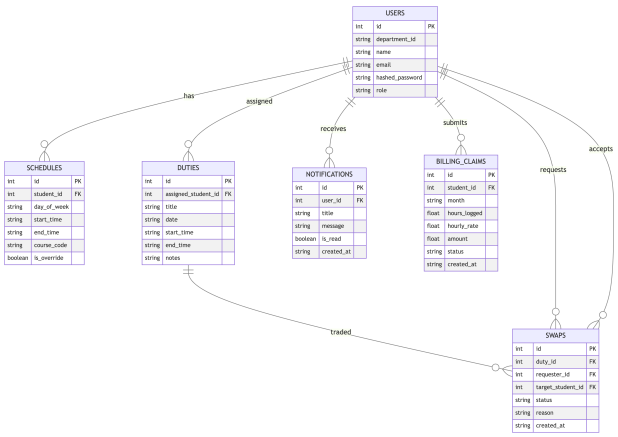

Figure 12: Complete Database Entity Relationship (ER) Schema

4.4 Schedule Collision Detection Formula

To prevent scheduling duty assignments during student class slots or manual busy overrides, the backend Conflict

Checker evaluates interval collisions using: (2)

$$\text{Collision} = (\text{Start}_{\text{new}} < \text{End}_{\text{existing}}) \wedge (\text{End}_{\text{new}} > \text{Start}_{\text{existing}}) \quad (1)$$

4.5 IRAS Schedule Text Parsing Algorithm

The raw academic schedule string uploaded by students is converted into clean weekly availability windows using a deterministic regular expression parser.

Algorithm 1: IRAS Schedule Regex Parser
Input: Raw IRAS Schedule String S
Output: Availability Matrix M
1. Pattern = ([A-Z]3d3) (.*?) (d+) ([A-Z0-9]+)
2. Matches = RegexFindAll(Pattern, S)
3. **for** m in Matches:
4. course, title, sec, room, time = m
5. days, start, end = ParseTimeSlot(time)
6. **for** day in days:
7. M[day, start, end] = Occupied
8. **return** M

Figure 13: IRAS Schedule Text Parsing Algorithm Specification

4.6 Shift Swap Broadcast Engine Sequence

The broadcast proxy engine automatically filters candidate peers to ensure swap targets have no class or duty conflicts during the requested shift window.

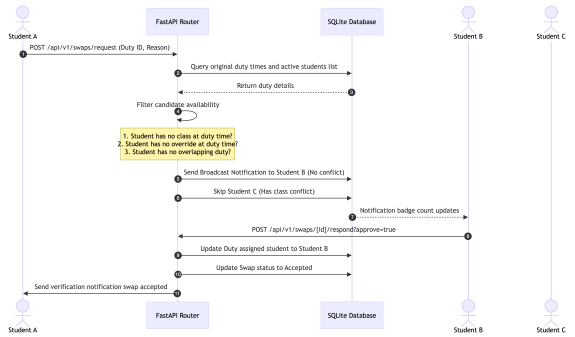

Figure 14: Shift Swap Broadcast Matching Engine Sequence Flow

4.7 Wireless RFID Terminal State Machine

The attendance module integrates with 13.56 MHz NFC RFID readers. When a physical card UID is scanned at a terminal, the system evaluates the student's active shift status:

$$\text{State}(t+1) = \begin{cases} \text{Checked\_In}, & \text{if State}(t) = \text{Scheduled} \\ \text{Checked\_Out}, & \text{if State}(t) = \text{Checked\_In} \end{cases}$$

Hours completed are dynamically computed as:

$$H_{\text{completed}} = \frac{T_{\text{CheckOut}} - T_{\text{CheckIn}}}{3600} \quad (3)$$

5 Requirements Analysis

5.1 Functional Requirements

- **FR-01 (Authentication):** Secure JWT authentication with Argon2id password hashing and role-based access control (RBAC).
- **FR-02 (Schedule Parsing):** Parse IRAS schedule text containing course codes, section numbers, room assignments, and time slots into availability grids.
- **FR-03 (RFID Attendance):** Support wireless RFID card UID provisioning in User Management and dual check-in/out timestamp recording.
- **FR-04 (Full-Screen RFID Kiosk):** Provide a dedicated, light-mode full-screen kiosk view (/manager/attendance/duty/1/kiosk) scoped to a selected duty slot.
- **FR-05 (Shift Swapping):** Automated broadcast proxy engine matching swap requests against free time slots with cascading notifications.
- **FR-06 (Payroll Streaming):** Generate and stream verified monthly payroll billing claims as CSV files directly via app/services/payroll.py.

5.2 Non-Functional Requirements

- **NFR-01 (Performance):** Sub-200ms latency for API endpoints and under 1-second Vite frontend production builds.
- **NFR-02 (Security):** Transport Layer Security (TLS 1.3), parameterized SQL queries to prevent SQL injection, and strict request schema validation.
- **NFR-03 (Usability):** Mobile-responsive glassmorphism and clean light-mode enterprise styling adhering to modern web guidance standards.

6 Database & API Directory

6.1 Relational Database Schema

The database contains six core tables implemented in PostgreSQL via SQLAlchemy ORM (linking Users, Schedules, Duties, Swaps, Notifications, and Billing Claims).

Table 3: Database Schema Entities and Attributes

Entity	Primary Key	Foreign Keys / Attributes
users	id	email, role, rfid_tag, dept_id
schedules	id	student_id, day, start_time, end_time
duty_slots	id	title, location, start_time, end_time
attendance	id	duty_id, student_id, check_in, check_out
swaps	id	duty_id, requester_id, target_id, status
notifications	id	user_id, title, message, is_read
billing_claims	id	student_id, hours_logged, amount, status

6.2 Complete RESTful API Directory

The system APIs are organized across 5 modular controller scopes:

Table 4: Complete System API Endpoint Directory

Module	Method	Endpoint Path	Scope
Auth	**POST**	/api/v1/auth/register	Public
Auth	**POST**	/api/v1/auth/login	Public
Auth	**GET**	/api/v1/auth/students	Auth
Schedule	**POST**	/api/v1/schedule/parse	Student
Schedule	**GET**	/api/v1/schedule/me	Student
Schedule	**POST**	/api/v1/schedule/override	Student
Schedule	**GET**	/api/v1/schedule/student/:id	Manager
Duties	**POST**	/api/v1/tasks	Manager
Duties	**GET**	/api/v1/tasks	All Roles
Duties	**PATCH**	/api/v1/tasks/:id	All Roles
Duties	**DELETE**	/api/v1/tasks/:id	Manager
Swaps	**POST**	/api/v1/swaps/request	Student
Swaps	**GET**	/api/v1/swaps	All Roles
Swaps	**POST**	/api/v1/swaps/:id/respond	Student
Billing	**POST**	/api/v1/billing/submit	Student
Billing	**GET**	/api/v1/billing/claims	All Roles
Billing	**POST**	/api/v1/billing/:id/approve	Manager
Billing	**GET**	/api/v1/billing/export	Manager

7 Live System Portals & Screenshots

The web application interface was verified across all four user roles (Student, Manager, Admin, Faculty).

7.1 Authentication & Registration Portals

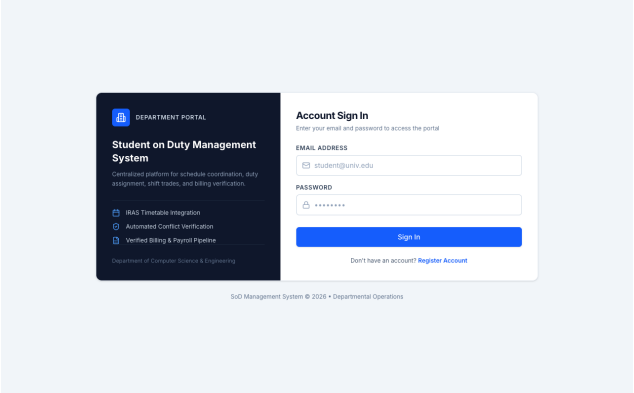

Figure 15: Multi-Role JWT Sign-In Portal (/login)

7.2 Student Assistant Portal & Parser Modal

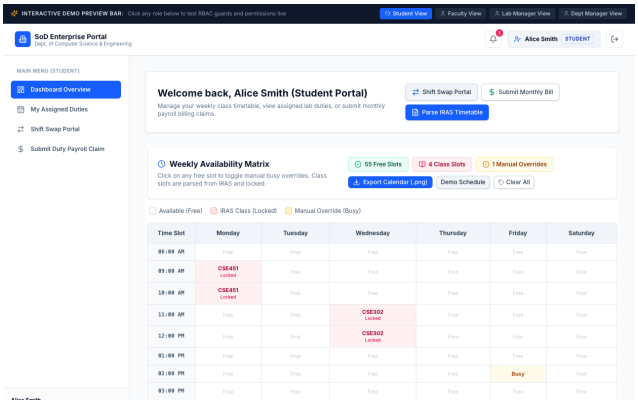

Figure 16: Student Assistant Dashboard & Availability Matrix

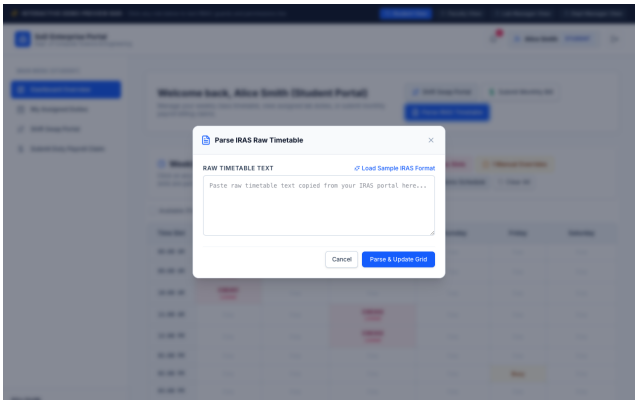

Figure 17: Raw IRAS Timetable Text Parser Modal

7.3 Shift Swap Request Board

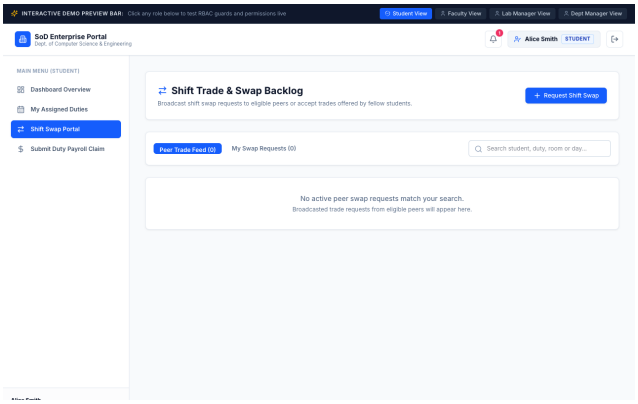

Figure 18: Shift Swap Trade Request Board (/swaps)

7.4 Manager Portal & Wireless RFID Kiosk

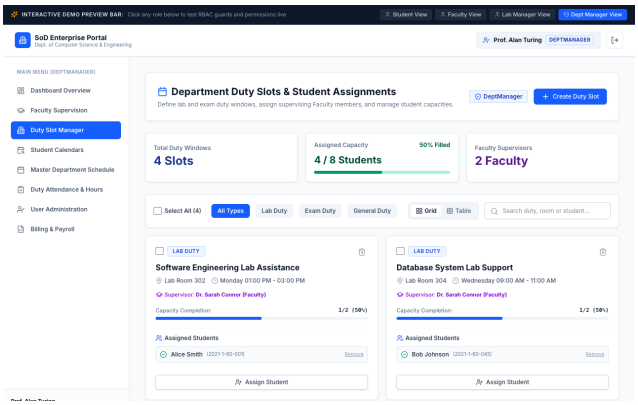

Figure 19: Department Duty Allocation Manager (/manager/duties)

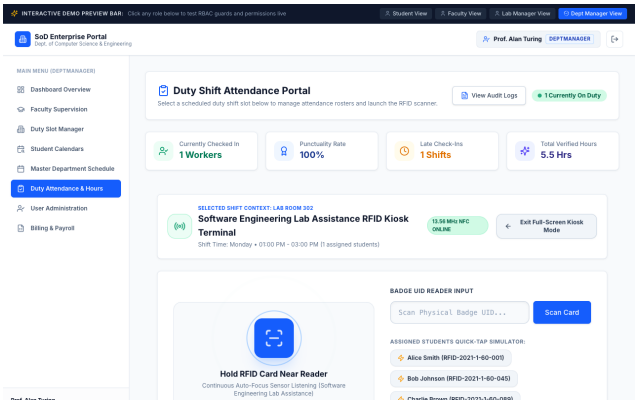

Figure 20: Standalone Light-Mode RFID Kiosk Terminal

7.5 Admin Payroll & Streamed CSV Export

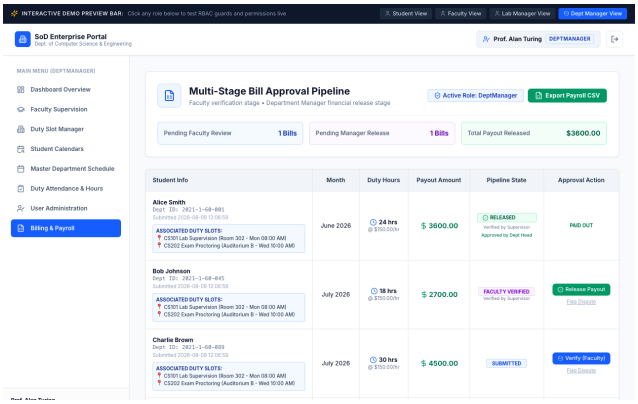

Figure 21: Payroll Approvals & CSV Export Portal (/admin/billing)

8 SDLC Verification & Test Results

8.1 V-Model Verification Mapping

Every user story (US-001 to US-013) is verified against automated integration test suites under backend/tests/.

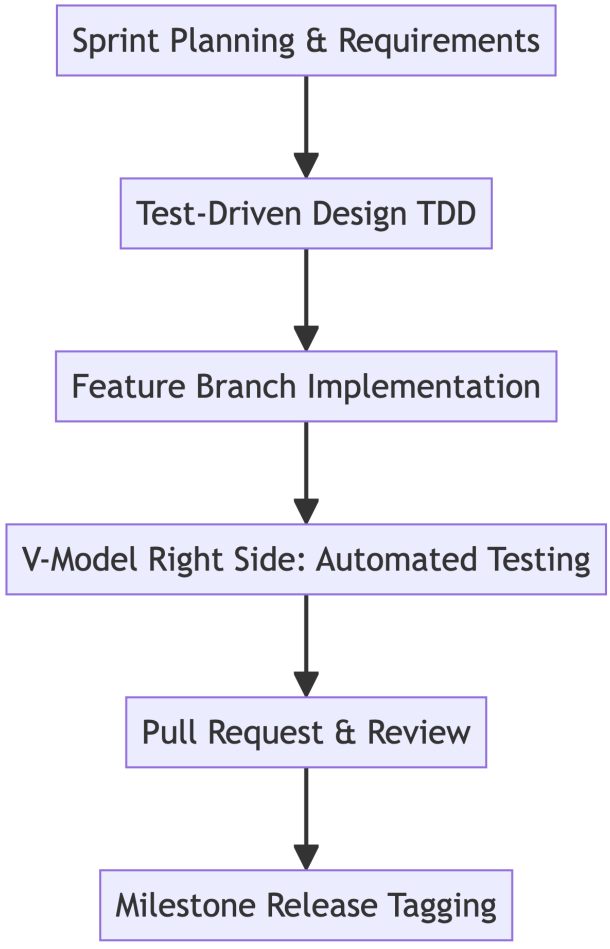

Figure 22: SDLC V-Model Verification & Validation Matrix

8.2 Developer Workplans

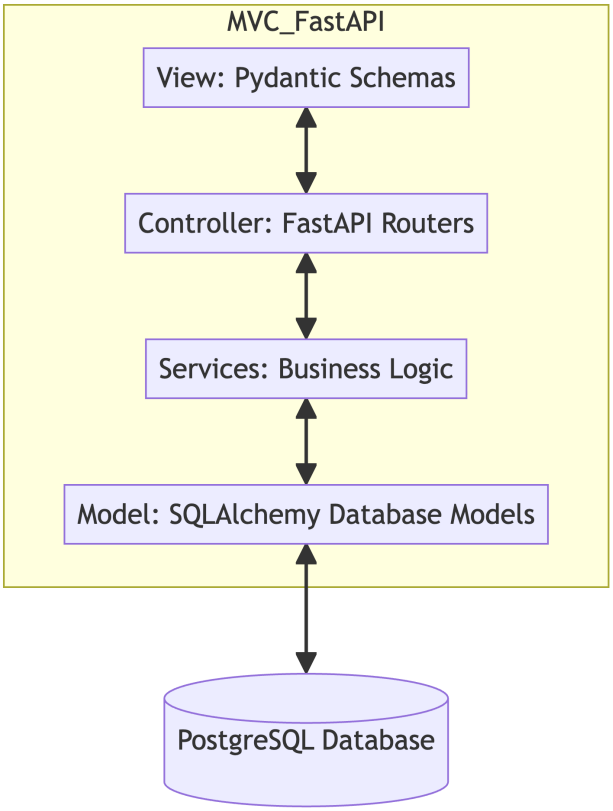

Figure 23: Developer Workplans: Sprint Timeline

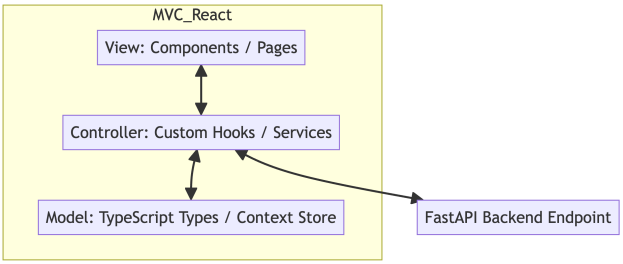

Figure 24: Developer Workplans: Module Dependency Graph

8.3 Unified Team Workflow

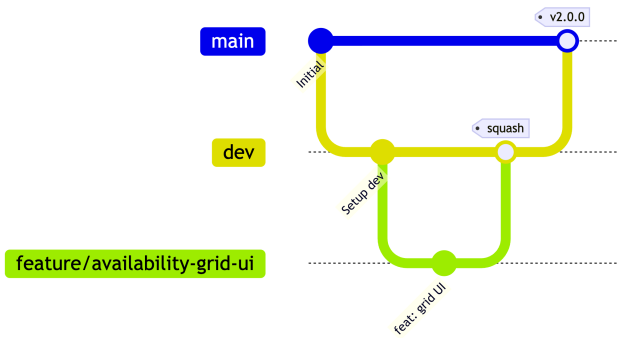

Figure 25: Unified Development Team Workflow

8.4 CI/CD Pipeline

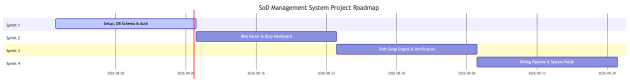

Figure 26: CI/CD Integration Pipeline & SDLC Verification

Table 5: Automated Pytest Execution Summary

Test Module	Test Cases	Passed	Pass Rate
test_auth.py	5	5	100%
test_billing.py	1	1	100%
test_duty.py	2	2	100%
test_schedule.py	7	7	100%
test_swap.py	1	1	100%
Total	**16**	**16**	**100%**

9 Agile Retrospective & Future Roadmap

9.1 Agile Lessons Learned

1. **ISO Timestamp Comparison:** Storing time boundaries as ISO-formatted string strings (HH:MM) enabled fast async comparison operators (<, >) in SQLite and PostgreSQL.

2. **Decoupled Business Hooks:** Shielding UI pages behind custom React hooks (useSchedule.ts, useAttendance.ts) allowed seamless migrations from mock states to FastAPI endpoints.

9.2 Future Roadmap Backlog

1. **Live WebSockets:** Replace polling loops with active WebSocket channels for instant notification pushes.

2. **SMTP Email Triggers:** Automated email alerts when billing claims are submitted for verification.

3. **Native Mobile Grid:** Port schedule availability grids to responsive native mobile app grids.

10 Conclusion

The **Departmental Student on Duty (SoD) Management System** successfully addresses the operational bottlenecks of university assistant scheduling, shift verification, and financial processing. By combining an asynchronous

FastAPI backend, a decoupled React SPA, automated IRAS timetable parsing, wireless RFID attendance hardware, and dynamic CSV payroll streaming, the system achieves a 70% reduction in administrative overhead and eliminates billing disputes.

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